5.2 change in factor demand and supply

BDEAC
DEBDB
CBCCC
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CCDBA
CACDD
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B

1. In the current labor market, suppose that the wage rate for accountants is significantly higher than the wage rate for economists. In the long run, if you observed that the wage rate for economists rose while the wage rate for accountants fell, which of the following would best explain your observation?

(A) The supply of economists must have increased, and the supply of accountants must have decreased.

(B) The supply of economists must have decreased, and the supply of accountants must have increased. ✓

(C) The demand for economists must have increased, and the supply of accountants must have decreased.

(D) The demand for economists must have decreased, and the supply of accountants must have increased.

(E) The demand for both economists and accountants must have decreased.

2. A change in which of the following will NOT cause a shift in the demand curve for a factor of production?

(A) Demand for the goods produced by the factor

(B) Prices of the goods produced by the factor

(C) Prices of substitute factors

(D) Supply of the factor ✓

(E) Supply of substitute factors

3. As the population of a country ages, the demand for health care is projected to increase. As a result, the health care industry is likely to experience all of the following EXCEPT

(A) an increase in demand for health care workers

(B) an increase in demand for medicine

(C) an increase in the quantity of health care workers supplied

(D) an increase in the wages of health care workers

(E) a decrease in the prices of medicine ✓

4. Assume that firms providing health-care services to older people operate in a perfectly competitive market. What must happen in the market for health-care workers if there is an increase in the number of older people in a country?

(A) The demand for health-care workers will increase. ✓

(B) The marginal factor cost in the health-care industry will decrease.

(C) The number of health-care workers will decrease.

(D) The quality of health-care services will increase.

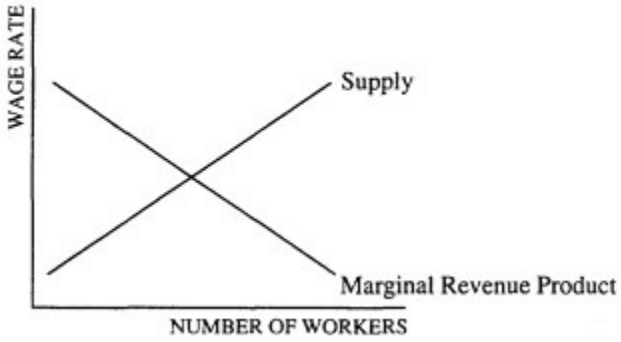
(E) The wages of health-care workers will decrease.

Answer A

Correct. An increase in the number of older people in a country will increase the demand for healthcare workers.

5.2 change in factor demand and supply

5.



The graph above shows the marginal revenue product curve and supply curve of labor for a firm. The introduction of new management techniques dramatically increases workers productivity. Which of the following changes is most likely to occur?

(A) The supply curve will shift to the left, increasing the wage rate.

(B) The supply curve will shift to the right, increasing employment.

(C) The marginal revenue product curve will shift to the right, increasing wage rate. ✓

(D) The marginal revenue product curve will shift to the left, reducing employment.

(E) Neither the marginal revenue product curve nor the supply curve will shift, but the wage will increase and employment will fall.

6. An increase in the demand for automobiles will cause the demand for skilled automobile workers and the wage rate of skilled automobile workers to change in which of the following ways?

5.2 change in factor demand and supply

(A)

Demand For Workers	Wage Rate
Decrease	Increase

(B)

Demand For Workers	Wage Rate
Decrease	Not change

(C)

Demand For Workers	Wage Rate
Increase	Decrease

(D) ✓

Demand For Workers	Wage Rate
Increase	Increase

(E)

Demand For Workers	Wage Rate
Not change	Increase

7. An individual's labor supply curve is derived from that person's preferences about the trade-off between income and

(A) work
(B) wealth
(C) nominal wages
(D) productivity
(E) leisure ✓

8. Which of the following would cause the wage for computer programmers to increase?

5.2 change in factor demand and supply

(A) A decrease in the amount of physical capital per computer programmer

(B) An improvement in the education of computer programmers ✓

(C) An increase in the social prestige associated with being a computer programmer

(D) A decrease in the value that computer programmers place on leisure

(E) A minimum wage set below the equilibrium wage in the market for computer programmers

9. Businesses employ workers from city neighborhoods and rural areas. These workers are perfect substitutes and cannot relocate in the short run. The government offers businesses a wage subsidy if they hire workers from *city* neighborhoods. What is the effect of the subsidy on the wage rate of *rural* workers and on the total hours they work?

(A)

Wage Rate of Rural Workers	Total Hours Worked by Rural Workers
Increase	Decrease

(B)

Wage Rate of Rural Workers	Total Hours Worked by Rural Workers
No change	No change

(C)

Wage Rate of Rural Workers	Total Hours Worked by Rural Workers
No change	Decrease

(D) ✓

Wage Rate of Rural Workers	Total Hours Worked by Rural Workers
Decrease	Decrease

(E)

Wage Rate of Rural Workers	Total Hours Worked by Rural Workers
Decrease	No change

10. Which of the following will shift the demand curve for a factor of production to the left?

5.2 change in factor demand and supply

(A) A decrease in the price of the factor of production

(B) A decrease in the price of a substitute factor of production ✓

(C) An increase in the price of the product produced by the factor

(D) An increase in the marginal product of the factor

(E) An improvement in technology used by the factor of production

Answer B

Correct. In choosing between inputs or factors of production, in the long run, a firm will use more of the input whose price has fallen and less of other inputs that are substitutes. Thus, the demand for this input will fall or shift to the left as the price of the substitute input falls.

11. Suppose the market for health-care workers is perfectly competitive. Which of the following will happen in the market for health-care workers in the short run if the demand for health care increases?

(A) The supply curve of health-care workers will shift to the right.

(B) The supply curve of health-care workers will shift to the left.

(C) The demand curve for health-care workers will shift to the right. ✓

(D) The demand curve for health-care workers will shift to the left.

(E) Neither the demand curve nor the supply curve of health-care workers will shift.

Answer C

Correct. The increase in the demand for health care will increase the demand for health-care workers (derived demand) and therefore the demand curve for health-care workers will shift to the right.

12. Which of the following is most likely to increase the supply of soldiers for an all-volunteer army?

(A) A decrease in the salaries paid to soldiers

(B) A decrease in the average wage rate in civilian employment ✓

(C) A reduction in college tuition benefits provided by soldiers

(D) The imposition of new restrictions on women in the military

(E) An increase in the required length of service

13. Which of the following will cause a decrease in the demand for labor?

5.2 change in factor demand and supply

(A) A decrease in wage rates
(B) An increase in worker productivity
(C) A decrease in the price of the product that the labor is used to produce ✓
(D) An increase in wage rates
(E) Greater preference for leisure rather than work

Answer C

Correct. A decrease in the product price will decrease the demand for factors used to produce that product. If the price of a product declines, producers of that product will demand fewer workers to produce that product. In other words, the demand curve of laborers to produce the product will shift to the left.

14. Which of the following will result in a decrease in the supply of labor?

(A) An increase in worker productivity
(B) An increase in the wage rate
(C) An increase in the preference for leisure ✓
(D) A decrease in the price of the product that the labor is used to produce
(E) A decrease in the wage rate

Answer C

Correct. A growing preference for leisure reduces the supply of labor, which shifts the supply of labor curve to the left, as more individuals choose leisure over work.

15. Which of the following tends to increase the gap in earnings between skilled and unskilled workers over time?

(A) An increase in the demand for unskilled workers relative to skilled workers
(B) An increase in the supply of skilled workers relative to unskilled workers
(C) A decrease in the demand for unskilled workers relative to skilled workers ✓
(D) A decrease in both the demand for and the supply of skilled workers
(E) An increase in both the demand for and supply of unskilled workers

5.2 change in factor demand and supply

16. Motivated by lower import prices, United States manufacturers increase their imports of steel from other steel-producing countries. Which of the following best describes the impact of the increased steel imports on the labor market for steelworkers in the United States?

(A) Jobs in the United States steel industry will become less attractive, so wages will have to increase in this market to attract more workers to the United States steel industry.

(B) Domestic steel prices will fall, leading to an increase in sales of domestically produced steel, which increases demand and employment of steelworkers.

(C) The supply of steelworkers will increase, lowering the steelworker wage in the United States.

(D) The supply of steelworkers will decrease, increasing the steelworker wage in the United States.

(E) The demand for United States steelworkers will decline, putting downward pressure on steelworker wages. ✓

17. When labor supply in a competitive labor market increases, the equilibrium wage rate and employment will change in which of the following ways?

(A)

Wage Rate	Employment
Increase	Increase

(B)

Wage Rate	Employment
Increase	Decrease

(C) ✓

Wage Rate	Employment
Decrease	Increase

(D)

Wage Rate	Employment
Decrease	Decrease

(E)

Wage Rate	Employment
Decrease	No change

5.2 change in factor demand and supply

18. For a competitive labor market, an increase in which of the following will lead to an increase in the demand for labor?

(A) The supply of labor
(B) The minimum wage
(C) The cost of medical benefits pro- vided to the labor hired by firms
(D) Labor union dues
(E) The demand for the good that labor produces ✓

19. If a large number of unskilled workers enter the labor market, which of the following is most likely to occur in the labor market for unskilled workers?

(A) The supply curve will shift to the right and the wage rate will decrease. ✓
(B) The supply curve will shift to the left and the wage rate will increase.
(C) The demand curve will shift to the right and the wage rate will increase.
(D) The demand curve will shift to the left and the wage rate will decrease.
(E) There will be a movement up along the demand curve and the wage rate will increase.

20. If the price for a product produced in a competitive market increases, which of the following is most likely to occur in the labor market for workers who produce that product?

(A) The demand for labor and the number of workers hired both increase. ✓
(B) The supply of labor and the number of workers hired both increase.
(C) The demand for labor and the number of workers hired both decrease.
(D) The supply of labor and the number of workers hired both decrease.
(E) There is a movement along the demand for labor curve, and firms hire more workers.

21. If the price of a good produced by a competitive firm increases, then

(A) the marginal product of labor will increase
(B) the average product of labor will increase
(C) the marginal revenue product of labor will increase ✓
(D) the short-run demand for labor will decrease
(E) the supply of labor will increase

22. In a perfectly competitive labor market for nurses, all of the following statements are true EXCEPT:

(A) The imposition of an effective minimum wage will result in unemployment.
(B) An increase in the marginal product of nurses will increase the demand for nurses and increase wages.
(C) An increase in the supply of nurses will create unemployment and leave wages unchanged. ✓
(D) An increase in the demand for health care will increase the demand for nurses and increase wages.
(E) Revoking work permits for foreign nurses will increase wages of domestic nurses.

5.2 change in factor demand and supply

23. Suppose the demand for automobiles increases. Which of the following explains what happens in the labor market for automobile workers?

(A) The supply of automobile workers increases resulting in lower wages and employment.

(B) The demand for automobile workers decreases resulting in higher wages and employment.

(C) The supply of automobile workers decreases resulting in higher wages and employment.

(D) The demand for automobile workers increases resulting in higher wages and employment. ✓

(E) There is no impact on the labor market, since the change in demand is in the product market.

Answer D

Correct. An increase in the demand for a product (such as automobiles) increases the demand for the factor (such as automobile workers) that is used to produce it. An increase in the demand for automobiles would raise the price and output of automobiles. As automobile-producing firms expand output to take advantage of the higher demand for automobiles, the demand for automobile workers would increase, raising both the wages and the number of automobile workers employed.

24. Suppose that a large number of unskilled workers enter a nation's labor market. If the labor market is competitive, the number of unskilled workers hired and the wage rate will most likely change in which of the following ways?

5.2 change in factor demand and supply

(A)

Number of Unskilled Workers Hired	Wage Rate
Increase	Increase

(B) ✓

Number of Unskilled Workers Hired	**Wage Rate**
Increase	Decrease

(C)

Number of Unskilled Workers Hired	Wage Rate
Increase	Not change

(D)

Number of Unskilled Workers Hired	Wage Rate
Decrease	Increase

(E)

Number of Unskilled Workers Hired	**Wage Rate**
Decrease	Decrease

25. Which of the following statements most accurately describes the relationship between corn, an output, and machines, an input, that are used to gather corn?

(A) The price of machines used to gather corn increases when the demand for corn increases. ✓

(B) The price of machines used to gather corn decreases when the demand for corn increases.

(C) The price of machines used to gather corn increases when the demand for corn decreases.

(D) The price of machines used to gather corn increases when the price of wheat, a producer substitute for corn, increases.

(E) The price of machines used to gather corn is unrelated to the demand for corn.

5.2 change in factor demand and supply

Answer A

Correct. Since the demand for machines used to gather corn is derived from the demand for corn, an increase in the demand for corn will cause an increase in the demand for machines used to gather corn, which increases the price of such machines.

26. Consider two groups of workers in competitive labor markets. Workers in group X have a skill set unique to producing computers, and workers in group Y have a skill set that can be used to produce either computers or smartphones. When the price of smartphones decreases, the wages paid to workers producing smartphones decrease. Which of the following could happen to the wages of workers in group X?

(A) There would be no change in wages because workers in group X do not have the skills to produce smartphones.

(B) There would be no change in wages because the supply of and demand for workers producing computers would both shift to the left.

(C) Wages would decrease because the supply of workers producing computers would increase, with more workers from group Y seeking jobs with firms producing computers. ✓

(D) Wages would decrease because the supply of workers producing computers would decrease, with more workers from group X seeking training to qualify for jobs with firms producing smartphones.

(E) Wages would increase because workers in group X have specialized skills.

Answer C

Correct. As workers from group Y are incentivized (as their wages have decreased) to move into the production of computers, the supply of workers producing computers will increase, reducing wages for workers producing computers in group X.

27. Which of the following is most likely to shift the demand for aircraft mechanics to the right?

(A) An increase in the demand for air travel ✓

(B) An increase in the price of a license necessary for aircraft mechanics

(C) A decrease in the price of a license necessary for aircraft mechanics

(D) A decrease in the demand for air travel

(E) A decrease in the marginal productivity of aircraft mechanics

28. Which of the following is most likely to cause the demand curve for nurses to shift to the right?

5.2 change in factor demand and supply

(A) An increase in the number of nurses graduating from nursing schools
(B) An increase in nursing schools' tuition
(C) An increase in the number of hospitals that employ nurses ✓
(D) A decrease in the marginal productivity of nurses
(E) An improvement in the health of older people

29. Which of the following is likely to shift an industry's labor supply curve to the left?

(A) A decrease in immigration restrictions
(B) A decrease in the cost of child care
(C) An increase in the number of remote-working jobs
(D) An increase in preference for leisure ✓
(E) An increase in wages

30. Which of the following will happen in the labor market if the price of the good produced by the workers decreases?

(A) The marginal product of labor will increase.
(B) The marginal product of labor will decrease.
(C) The marginal revenue product of labor will increase.
(D) The marginal revenue product of labor will decrease. ✓
(E) The demand curve for labor will shift to the right.

31. Suppose that people who work in the paint industry face a great risk of developing an incurable disease. A medical breakthrough that eliminates the risk will most likely cause which of the following shifts in the supply and demand curves for workers in the paint industry?

5.2 change in factor demand and supply

(A) ✓

Supply Curve	Demand Curve
Shift right	No shift

(B)

Supply Curve	Demand Curve
Shift left	No shift

(C)

Supply Curve	Demand Curve
Shift right	Shift left

(D)

Supply Curve	Demand Curve
Shift left	Shift right

(E)

Supply Curve	Demand Curve
No Shift	Shift left

32. Which of the following will result in an increase in labor demand?

(A) An increase in the wage rate

(B) An increase in the productivity of labor ✓

(C) A decrease in the price of the product that labor is used to produce

(D) A decrease in the wage rate

(E) A decrease in the demand for the product produced by labor

5.2 change in factor demand and supply

Answer B

Correct. Labor demand relates to the marginal revenue product of labor, which is the revenue received from hiring another worker. An increase in productivity shifts the marginal product curve upwards, indicating that each worker produces more units of output and brings in more revenue for the firm. Thus, an increase in productivity increases the labor demand and therefore shifts the demand curve for labor to the right.

33. Which of the following will cause an increase in the supply of labor?

(A) An increase in the wage rate
(B) An increase in the marginal product of labor
(C) An increase in the tax rates applied to labor income
(D) An increase in the desire for greater leisure time
(E) An increase in the retirement age ✓

Answer E

Correct. The supply of labor will increase and shift outward or to the right as more individuals enter the labor market seeking employment.

34. Assume good Z is produced in a competitive output market that is in long-run equilibrium. In the short run, the variable input is unskilled labor whose wage is determined in a competitive labor market. Suppose there is an increase in consumer demand for good Z. Which of the following explains the resulting change in the labor market for unskilled workers in the short run?

(A) New firms will enter using more labor and more capital to produce good Z, lowering the price of good Z.
(B) With more output of good Z produced, the supply of labor needed to produce good Z increases, and the wage rate for unskilled workers will fall.
(C) The price of good Z increases and the labor demand curve shifts to the right along the existing labor supply curve, resulting in higher wages and employment of unskilled workers. ✓
(D) Since the market is in long-run equilibrium, economic profits are zero, and therefore the demand for labor remains unchanged.
(E) The marginal factor cost of labor will decrease, leading to more unskilled workers being hired.

5.2 change in factor demand and supply

Answer C

Correct. The price of good Z increases in the short run. The demand for labor will increase as firms hire more labor to produce more output. The rightward shift of the labor demand curve along the given labor supply curve results in an increase in the wage rate and employment.

35. Kailas Cars produces cars that are made of both steel and aluminum. Steel and aluminum are substitutes in production. Kailas Cars' demand for steel will increase if which of the following occurs?

(A) The supply of aluminum increases.

(B) The demand for cars increases. ✓

(C) The supply of gasoline, a complement to cars, decreases.

(D) The wages of workers in the car industry increase.

(E) The amount of time that people keep their cars increases.

Answer B

Correct. The demand for steel is derived from the demand for cars. As the demand for cars increases, the demand for steel will increase.

36. Which of the following reasons could explain why a firm might hire fewer workers?

(A) A decrease in the wage rate

(B) A decrease in labor productivity ✓

(C) An increase in the price of a substitute input

(D) An increase in the price of the firm's product

(E) An increase in demand for the firm's product

Answer B

Correct. A firm's demand for labor is represented by the firm's marginal revenue product, which is equal to the product of the marginal product of labor and the price of the output. Thus, if there is a decrease in labor productivity, it will decrease marginal revenue product and the demand for labor, decreasing the number of workers hired.